# SIMPLE OVERVIEW OF CONFORMAL PREDICTION

Andreas Koukorinis

# WHY PREDICTION INTERVALS?

- Modern ML returns point predictions $\hat{f}(x)$ with no quantified uncertainty
- Bootstrap, Bayesian posteriors, quantile regression: all need model assumptions or asymptotics
- Conformal prediction (Vovk–Shafer–Romano) wraps any black box and returns a set $\mathcal{C}(x)$ such that

$$\mathbb{P}\Big(Y_{n+1} \in \mathcal{C}(X_{n+1})\Big) \geq 1 - \alpha$$

- Distribution-free, finite-sample, model-agnostic — the only assumption is exchangeability of $(X_i, Y_i)_{i=1}^{n+1}$
- Trade interval/set size for coverage; user picks the miscoverage level $\alpha$

## THE CONFORMAL RECIPE (ANGELOPOULOS & BATES 2022)

Given a fitted predictor $\hat{f}$, a held-out calibration set $\{(X_i, Y_i)\}_{i=1}^n$, and level $\alpha$:

1. Define a nonconformity score $s(x, y) \in \mathbb{R}$: "how surprising is $(x, y)$?"
   - Regression: $s(x, y) = |y - \hat{f}(x)|$
   - Classification: $s(x, y) = 1 - \hat{f}_y(x)$ (one minus softmax of true class)
2. Compute calibration scores $s_i = s(X_i, Y_i)$ for $i = 1, \ldots, n$
3. Take the empirical quantile

$$\hat{q} = \text{Quantile}\left(\{s_i\}_{i=1}^n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$$

4. Predict the set $\mathcal{C}(X_{n+1}) = \{y : s(X_{n+1}, y) \le \hat{q}\}$

**Theorem (Marginal coverage)**

*If $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable, then $\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \ge 1 - \alpha$.*

# VARIANTS: SCORES THAT BUY YOU ADAPTIVITY

- APS (Adaptive Prediction Sets): rank softmax probabilities, cumulate until mass $\geq \hat{q}$ — sets grow on hard inputs, shrink on easy ones
- RAPS: regularise APS to prevent pathological sets on tail classes
- CQR (Conformalised Quantile Regression, Romano et al.):

$$s(x, y) = \max\left(\hat{q}_{\alpha/2}(x) - y,\ y - \hat{q}_{1-\alpha/2}(x)\right)$$

  **intervals adapt to heteroscedasticity** — asymmetric, locally varying width
- Conformal Risk Control: replace coverage with any monotone loss $\ell(\lambda)$; tune threshold $\lambda$ so $\mathbb{E}[\ell] \leq \alpha$
- Learn-Then-Test: same for non-monotone risks via multiple testing
- All free choices — the coverage guarantee survives regardless of score

## BEYOND SPLIT CP: DATA-EFFICIENT VARIANTS

Split CP wastes data: half (or more) reserved for calibration.

- Full Conformal: retrain $\hat{f}$ for every candidate $y$ — exact, but $O(|\mathcal{Y}|)$ fits per test point
- Cross-Conformal / CV+: $K$-fold variant; cheap and statistically efficient
- Jackknife+ (Barber, Candès, Ramdas, Tibshirani):

$$\mathbb{P}\left(Y_{n+1} \in \mathcal{C}^{\mathrm{J+}}(X_{n+1})\right) \geq 1 - 2\alpha$$

  uses leave-one-out residuals; distribution-free guarantee at $2\alpha$ price
- Jackknife+ after Bootstrap (J+aB): the same idea wrapped around a bootstrap ensemble — no leave-one-out refit needed
- Rule of thumb:
  - **Lots of data, cheap model** $\Rightarrow$ split CP
  - **Scarce data, expensive model** $\Rightarrow$ Jackknife+ / cross-conformal

# TIME SERIES: EXCHANGEABILITY BREAKS

Stocker et al. (2025) organise CP-for-time-series into four families:

1. Reweight calibration — weighted conformal prediction (WCP); recency weights, covariate-shift weights
2. Refresh residuals via OOB bootstrap
   - EnbPI (Xu & Xie 2023): bootstrap-ensemble residuals; no data splitting, no retraining at test time; asymptotic conditional + marginal coverage under $\beta$-mixing
   - SPCI: sequential online refinement of EnbPI
3. Adapt $\alpha$ online: ACI updates the nominal level after each observation

$$\alpha_{t+1} = \alpha_t + \gamma\left(\alpha - \mathbb{1}\{Y_t \notin \mathcal{C}_t\}\right)$$

   variants: AgACI (online aggregation), Conformal PID, MultiValid
4. Block randomisation: block conformal prediction (BCP) over contiguous chunks

## WHAT COVERAGE REALLY MEANS

- Marginal coverage (what the theorem gives):

$$\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha \quad \text{averaged over } X$$

- Conditional coverage (what you actually want):

$$\mathbb{P}(Y \in \mathcal{C}(X) \mid X = x) \geq 1 - \alpha \quad \forall x$$

- Vovk impossibility: distribution-free conditional coverage is impossible without restrictive assumptions; achievable only with trivial (full-support) sets
- Practical relaxations (Dieuleveut & Zaffran 2025):
  - Class-conditional / group-conditional CP: coverage per group
  - Mondrian CP: partition $\mathcal{X}$ into taxonomic cells, calibrate per cell
  - Calibration-conditional (PAC-style) bounds: coverage that holds w.p. $1 - \delta$ over

## TAKE-HOME AND READING ORDER

- CP is a wrapper, not a model — drops on top of any predictor
- Pick the variant by your data regime:
  - **IID, lots of data** $\Rightarrow$ Split CP
  - **IID, scarce data** $\Rightarrow$ Jackknife+ or CV+
  - **Time series, stationary** $\Rightarrow$ EnbPI
  - **Time series, drift** $\Rightarrow$ ACI / AgACI / Conformal PID
  - **Asymmetric / heteroscedastic regression** $\Rightarrow$ CQR
- Libraries: `MAPIE`, `crepes`, `conformal-prediction` (Angelopoulos)
- Reading order:
  1. Angelopoulos & Bates (2022) – *A Gentle Introduction to Conformal Prediction*
  2. Dieuleveut & Zaffran (2025) – Hi! PARIS tutorial deck
  3. Stocker et al. (2025) – *A Gentle Introduction to Conformal Time Series Forecasting*
  4. Xu & Xie (2023) – *Conformal Prediction for Time Series* (EnbPI)